

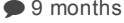
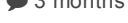
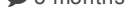
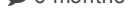
Problems

Rosalind is a platform for learning bioinformatics and programming through problem solving. [Take a tour](#) to get the hang of how Rosalind works.

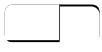
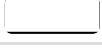
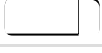
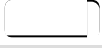
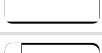
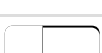
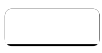
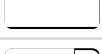
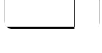
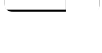
Last win: 董偉業 [tung wai yip](#) vs. ["Find a Median String"](#), 7 minutes ago

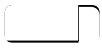
Problems: 258 (total), users: 30357, attempts: 516980, correct: 297467

ID	Title	Solved By	Correct Ratio	Questions	Solutions	Explanation
BA1A	Compute the Number of Times a Pattern Appears in a Text	58			1 day	
BA1B	Find the Most Frequent Words in a String	214			1 week	
BA1C	Find the Reverse Complement of a String	190			1 week	
BA1D	Find All Occurrences of a Pattern in a String	167			1 week	
BA1F	Find a Position in a Genome Minimizing the Skew	122			1 week	
BA1G	Compute the Hamming Distance Between Two Strings	57			1 week	
BA1H	Find All Approximate Occurrences of a Pattern in a String	115			1 week	
BA1I	Find the Most Frequent Words with Mismatches in a String	79			1 week	
BA1J	Find Frequent Words with Mismatches and Reverse Complements	70			1 week	
BA1K	Generate the Frequency Array of a Strings	76			1 week	
BA1L	Implement PatternToNumber	76		1 week	1 week	
BA1M	Implement NumberToPattern	72			1 week	
BA1N	Generate the d-Neighborhood of a String	63			1 week	
BA2A	Implement MotifEnumeration	52			1 week	
BA2B	Find a Median String	54			21 hours	
BA2C	Find a Profile-most Probable k-mer in a String	52			1 week	
BA2H	Implement DistanceBetweenPatternAndStrings	38		1 month	1 week	
BA3A	Generate the k-mer Composition of a String	55			1 week	
BA4C	Generate the Theoretical Spectrum of	30			1 month	

	a Cyclic Peptide					
BA5A	Find the Minimum Number of Coins Needed to Make Change	232			 1 month	
BA5C	Find a Longest Common Subsequence of Two Strings	218		 4 months	 1 month	
BA5E	Find a Highest-Scoring Alignment of Two Strings	163		 8 months	 1 month	
BA5F	Find a Highest-Scoring Local Alignment of Two Strings	144			 9 months	
BA5G	Compute the Edit Distance Between Two Strings	191			 1 month	
BA5H	Find a Highest-Scoring Fitting Alignment of Two Strings	114		 1 year	 9 months	
BA5I	Find a Highest-Scoring Overlap Alignment of Two Strings	122			 9 months	
BA5K	Find a Middle Edge in an Alignment Graph in Linear Space	79		 3 months	 3 months	
BA5L	Align Two Strings Using Linear Space	66			 1 year	
BA6B	Compute the Number of Breakpoints in a Permutation	155		 3 months	 3 months	
BA9A	Construct a Trie from a Collection of Patterns	124		 6 months	 3 months	
BA9B	Implement TrieMatching	114			 5 months	
BA9C	Construct the Suffix Tree of a String	83			 6 months	
BA9D	Find the Longest Repeat in a String	93			 2 months	
BA9G	Construct the Suffix Array of a String	82			 4 months	
BA9I	Construct the Burrows-Wheeler Transform of a String	99			 1 year	
BA9J	Reconstruct a String from its Burrows-Wheeler Transform	90			 5 months	
BA9N	Find All Occurrences of a Collection of Patterns in a String	77			 5 months	
BA9O	Find All Approximate Occurrences of a Collection of Patterns in a String	62			 5 months	
BA9Q	Construct the Partial Suffix Array of a String	52		 3 months	 3 months	
NEWBA3C	Construct the Overlap Graph of a Collection of k-mers	32		 1 day	 1 month	
NEWBA3D	Construct the De Bruijn Graph of a String	36		 4 days	 1 week	
NEWBA3E	Construct the De Bruijn Graph of a Collection of k-mers	32		 4 days	 1 month	

NEWBA3F	Find an Eulerian Cycle in a Graph	24			 1 month	
NEWBA3G	Find an Eulerian Path in a Graph	25			 1 month	
NEWBA3H	Reconstruct a String from its k-mer Composition	24			 1 month	
NEWBA3I	Find a k-Universal Circular String	22			 1 month	
NEWBA4M	Solve the Turnpike Problem	15			 4 weeks	
BA1E	Find Patterns Forming Clumps in a String	123				
BA2D	Implement GreedyMotifSearch	37				
BA2E	Implement GreedyMotifSearch with Pseudocounts	37				
BA2F	Implement RandomizedMotifSearch	35				
BA2G	Implement GibbsSampler	28				
BA3L	Construct a String Spelled by a Gapped Genome Path	10				
BA3M	Generate All Maximal Non-Branching Paths in a Graph	7				
BA4A	Translate an RNA String into an Amino Acid String	43				
BA4B	Find Substrings of a Genome Encoding a Given Amino Acid String	33				
BA4D	Compute the Number of Peptides of Given Total Mass	21				
BA4E	Find a Cyclic Peptide with Theoretical Spectrum Matching an Ideal Spectrum	25				
BA4J	Generate the Theoretical Spectrum of a Linear Peptide	9				
BA4L	Trim a Peptide Leaderboard	7				
BA5B	Find the Length of a Longest Path in a Manhattan-like Grid	193		 3 months		
BA5D	Find the Longest Path in a DAG	143				
BA5J	Align Two Strings Using Affine Gap Penalties	104				
BA5M	Find a Highest-Scoring Multiple Sequence Alignment	89				
BA5N	Find a Topological Ordering of a DAG	10				
BA6A	Implement GreedySorting to Sort a Permutation by Reversals	154				
BA6C	Compute the 2-Break Distance Between a Pair of Genomes	118				

BA6D	Find a Shortest Transformation of One Genome into Another by 2-Breaks	4				
BA6E	Find All Shared k-mers of a Pair of Strings	120				
BA6F	Implement ChromosomeToCycle	8				
BA6G	Implement CycleToChromosome	8				
BA6H	Implement ColoredEdges	7				
BA6I	Implement GraphToGenome	6				
BA6J	Implement 2-BreakOnGenomeGraph	6				
BA6K	Implement 2-BreakOnGenome	5				
BA7A	Compute Distances Between Leaves	6				
BA7B	Compute Limb Lengths in a Tree	6				
BA7C	Implement AdditivePhylogeny	4				
BA7D	Implement UPGMA	5				
BA7E	Implement the Neighbor Joining Algorithm	5				
BA7F	Implement SmallParsimony	5				
BA8A	Implement FarthestFirstTraversal	4				
BA8B	Compute the Squared Error Distortion	4				
BA8C	Implement the Lloyd Algorithm for k-Means Clustering	4				
BA8D	Implement the Soft k-Means Clustering Algorithm	4				
BA8E	Implement Hierarchical Clustering	4				
BA9E	Find the Longest Substring Shared by Two Strings	90				
BA9F	Find the Shortest Non-Shared Substring of Two Strings	71				
BA9H	Pattern Matching with the Suffix Array	0				
BA9L	Implement BWMatching	70				
BA9M	Implement BetterBWMatching	66				
BA9P	Implement TreeColoring	3				
BA9R	Construct a Suffix Tree from a Suffix Array	56				
NEWBA3J	Reconstruct a String from its Paired Composition	16				
NEWBA3K	Generate Contigs from a Collection of Reads	16				

NEWBA4G	Implement LeaderboardCyclopeptideSequencing	20				
NEWBA4H	Generate the Convolution of a Spectrum	20				
NEWBA4I	Implement ConvolutionCyclopeptideSequencing	16				